CONCEPT DRAWING SET

JBL Synthesis Theater, 24 × 18

Dolby Atmos 7.2.4 dedicated theater with two rows of seating and an acoustically transparent 120″ screen. Drawn for the **Better** package (Reference in-wall LCR, separates, four in-wall subs); all three packages are compared on AV-701.

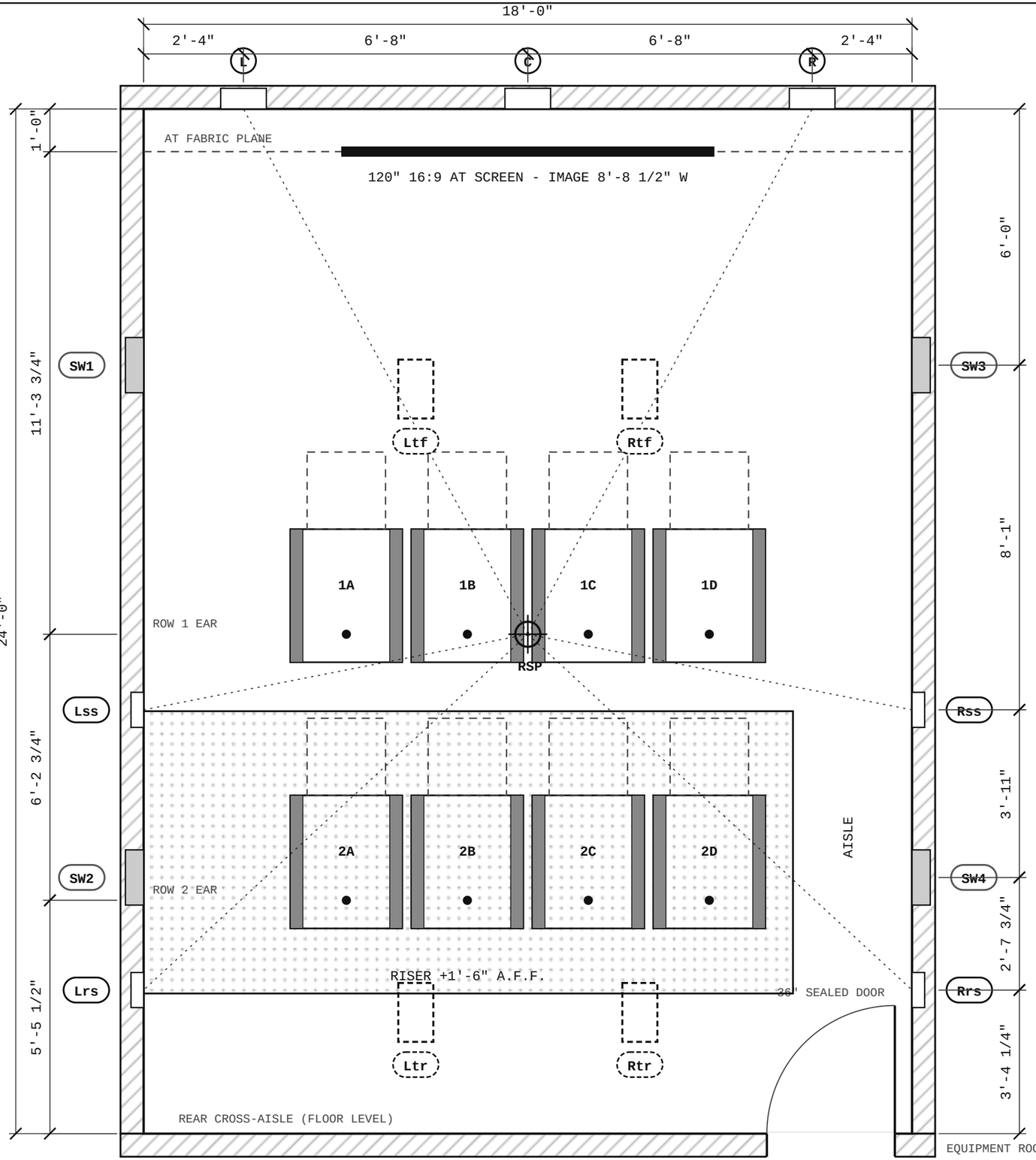
ROOM **24′ × 18′ × 9′ (110 m³)**	PACKAGE **Better: SCL-2 / SCL-3 / SCL-3**	FORMAT **Dolby Atmos 7.2.4, 4 × SSW-3**	SEATING **2 rows × 4, riser 1'-6"**	SCREEN **120″ 16:9 AT, 8'-8 1/2" wide**
ROW 1 VIEW **42° horizontal**	PROCESSOR **SDP-60, 13/16 outputs**	EST. LEVEL **106-109 dB peaks**	BASS SPREAD **3.0 dB mean seat SD**	EQUIPMENT LIST *use Show prices*

SHEET INDEX

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A-102	Reflected ceiling plan
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A-202	Side wall elevations
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AV-501	Schedules
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AV-602	Subwoofer planner (interactive)
AV-701	Good / Better / Best options
AV-702	Budget builder
V-801	3D model and client spin

DESIGN BASIS AND ASSUMPTIONS

- Clean-room design using published manufacturer data and the CHT calculation catalog (Phase 3). Formula IDs on AV-601.
- Room 24′ L × 18′ W; ceiling 9′ is an assumption. Finished-face dimensions to be field-verified.
- Screen on the 18′ wall. Full-width acoustically transparent front wall; L/C/R in-wall behind it.
- Equipment rack outside the theater beyond the back wall; door on the back wall at the right aisle.
- Placement windows from Dolby home-theater guidance as commonly cited; verify against the current Dolby document.
- Output figures are estimates (±3 dB). Speaker footprints marked ASSUMED in the catalog must be verified with JBL.
- Products and SKUs from the Savant integrator price list 20261103; Good / Better / Best compared on AV-701.



LEGEND

- In-wall loudspeaker (JBL SCL-2 / SCL-3)
- Subwoofer (JBL SSW-3)
- Ceiling loudspeaker (above)
- Recliner, upright (footrest dashed)
- Riser platform
- Reference ray from RSP
- Reference seating position (RSP)

SPEAKER ANGLES FROM RSP

TAG	MODEL	AZ	EL	DIST	OK
L	SCL-2	28.4°	1.6°	14'-0"	3/8
C	SCL-2	0.0°	1.8°	12'-3 3/4"	-
R	SCL-2	28.4°	1.6°	14'-0"	3/8
Lss	SCL-3	101.1°	16.9°	9'-7"	4/8
Rss	SCL-3	101.1°	16.9°	9'-7"	4/8
Lrs	SCL-3	132.8°	12.8°	12'-7"	2/8
Rrs	SCL-3	132.8°	12.8°	12'-7"	2/8
Ltf	SCL-3	24.6°	41.3°	8'-5"	4/8
Rtf	SCL-3	24.6°	41.3°	8'-5"	4/8
Ltr	SCL-3	163.5°	31.0°	10'-9 1/4"	6/8
Rtr	SCL-3	163.5°	31.0°	10'-9 1/4"	6/8

PLAN NOTES

1. Dimensions to finished interior faces. Concept dimensions; field-verify before framing.
2. Front wall is a full-width acoustically transparent (AT) fabric plane; L/C/R are in-wall behind it.
3. Check framing depth for in-wall SCL-2 and SSW-3 (2x6 walls recommended; see manufacturer templates).
4. Speaker centres are shown; see AV-501 for X/Y/Z coordinates and AV-601 for seat compliance.
5. Ear points (dots) at 1.05 m above row floor; RSP is row-1 centre between seats 1B/1C.
6. OK = seats (of 8) inside the Dolby placement window for that channel.

CHT

CHICAGOLAND HOME THEATER

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
B (concept, Savant-priced packages)

DATE
2026-11-09

DRAWN
CHT design engine v0.1

SCALE
3/8" = 1'-0" (on 11x17)

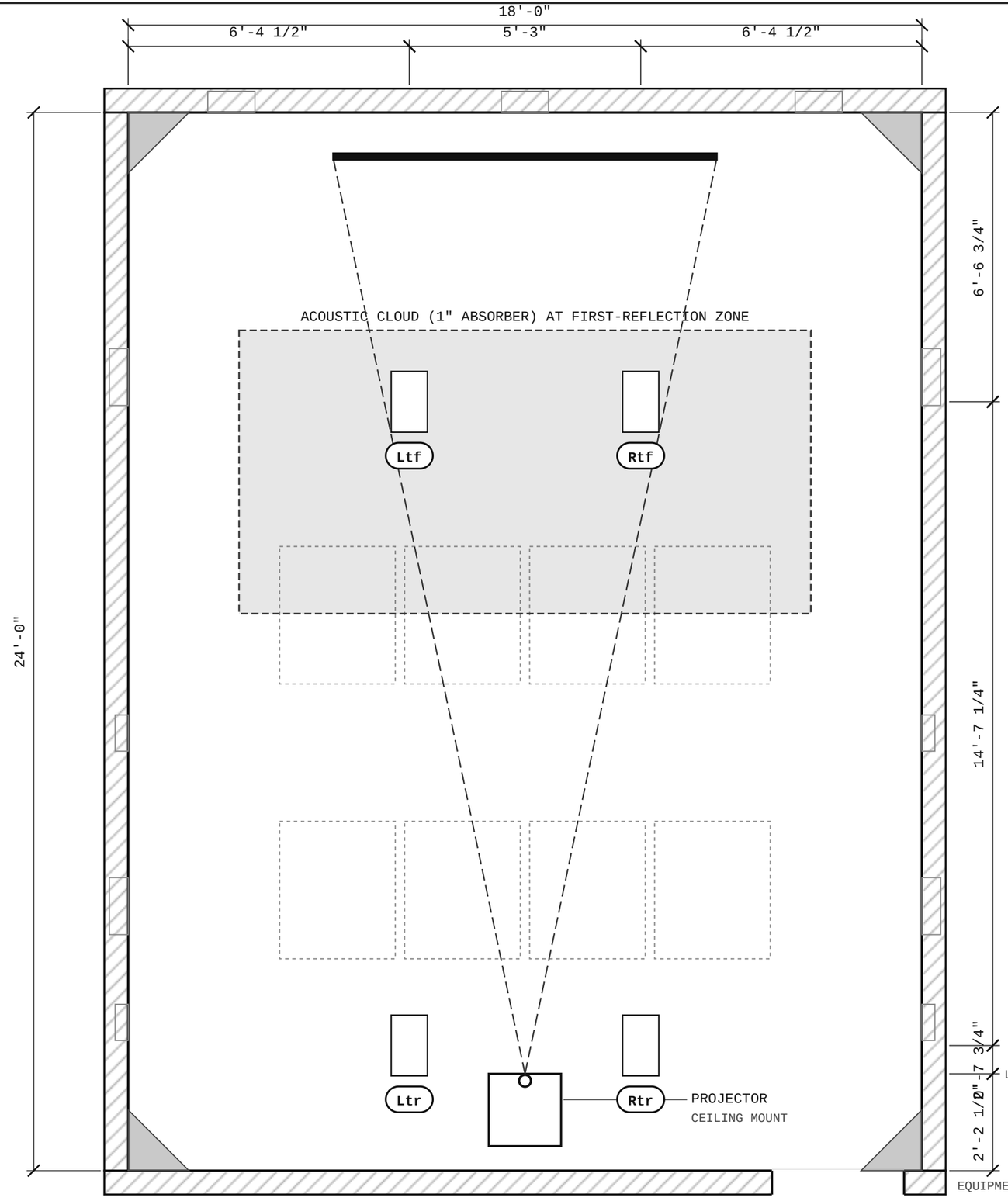
CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE

Floor Plan and Speaker Layout

A-101



LEGEND

- Ceiling loudspeaker (JBL SCL-3)
- Projector and mount (lens at front edge)
- Projected image edges
- Acoustic cloud / first-reflection zone
- Corner bass trap below (full height)
- Seats below (reference)

OVERHEAD SPEAKERS

TAG	X (L WALL)	Y (SCRN WALL)	EL@RSP
Ltf	6'-4 1/2"	6'-6 3/4"	41.3°
Rtf	11'-7 1/2"	6'-6 3/4"	41.3°
Ltr	6'-4 1/2"	21'-2"	31.0°
Rtr	11'-7 1/2"	21'-2"	31.0°

CEILING NOTES

1. Ceiling height 9'-0" (ASSUMED). Confirm joist direction and depth before locating ceiling speakers.
2. Projector lens at 20'-9 1/2" from screen plane, lens height 8'-4 1/4"; throw ratio 2.39:1.
3. Vertical lens shift required 79% of image height (published limit +/-85%; verify Sony definition).
4. Overhead speakers aimed toward midpoint between rows where the model allows rotation.
5. Provide a dedicated 20 A circuit and conduit (HDMI fibre + control) to the projector location.

CHT

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DRAWN
CHT design engine v0.1

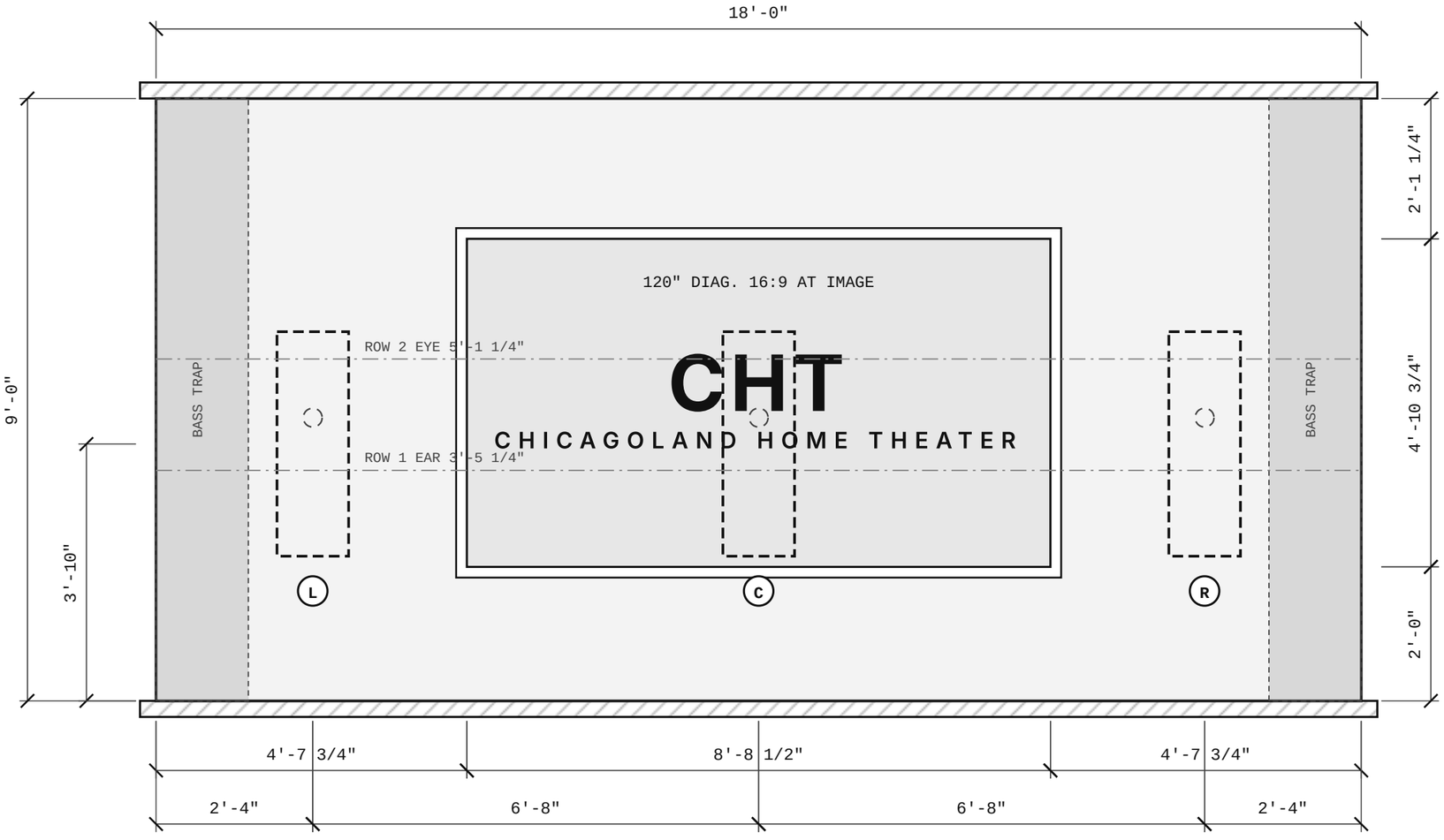
SCALE
3/8" = 1'-0" (on 11x17)

CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE
Reflected Ceiling Plan

A-102



ELEVATION NOTES

1. Image 8'-8 1/2" W x 4'-10 3/4" H; bottom of image 2'-0" A.F.F.; top 6'-10 3/4".
2. L/C/R JBL SCL-2 behind AT fabric, centred at 1.17 m (3'-10"); verify grille clear of screen frame.
3. Black, absorptive treatment behind the full AT plane; corner bass traps behind fabric.
4. Front-row vertical angle to top of image 16.3 deg (guidance ~15 deg, verify).

CHT

CHICAGOLAND HOME THEATER

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
B (concept, Savant-priced packages)

DATE
2026-11-09

DRAWN
CHT design engine v0.1

SCALE
1/2" = 1'-0" (on 11x17)

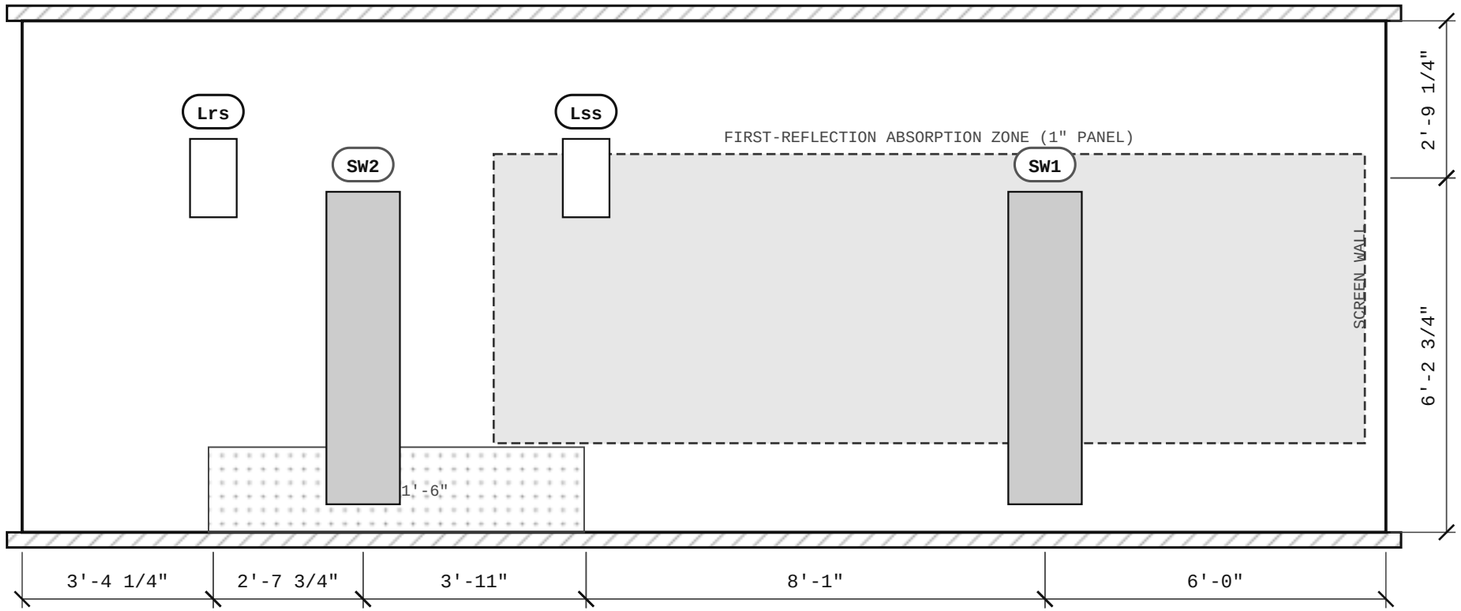
CONCEPT DESIGN

NOT FOR CONSTRUCTION

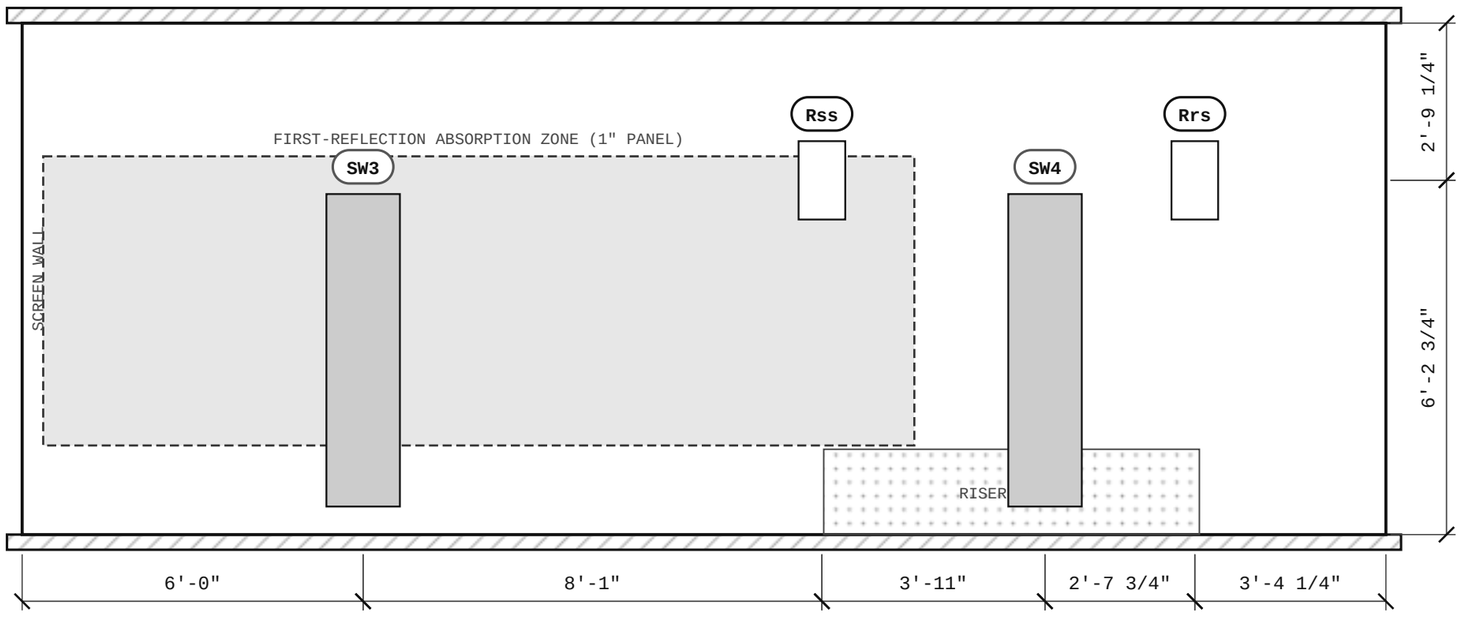
SHEET TITLE
Screen Wall Elevation

A-201

LEFT WALL - VIEWED FROM INSIDE THE ROOM



RIGHT WALL - VIEWED FROM INSIDE THE ROOM



NOTES

1. Surrounds JBL SCL-3 in-wall at a common centre height to clear row-2 heads on the riser.
2. SSW-3 subwoofers vertical, bottom of grille 6" A.F.F.; locate between studs per JBL template. Layout: Side walls at 1/4 and 3/4 length (see AV-602).
3. Absorption zones derived from first-reflection points of L/C/R for all eight seats, plus margin.
4. Treat the rear half of side walls with diffusion or leave reflective to avoid over-damping (see AV-601 RT60).
5. Subwoofer grilles may sit behind fabric; keep panels clear of sub cut-outs.

CHT

CHICAGOLAND HOME THEATER

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
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DRAWN
CHT design engine v0.1

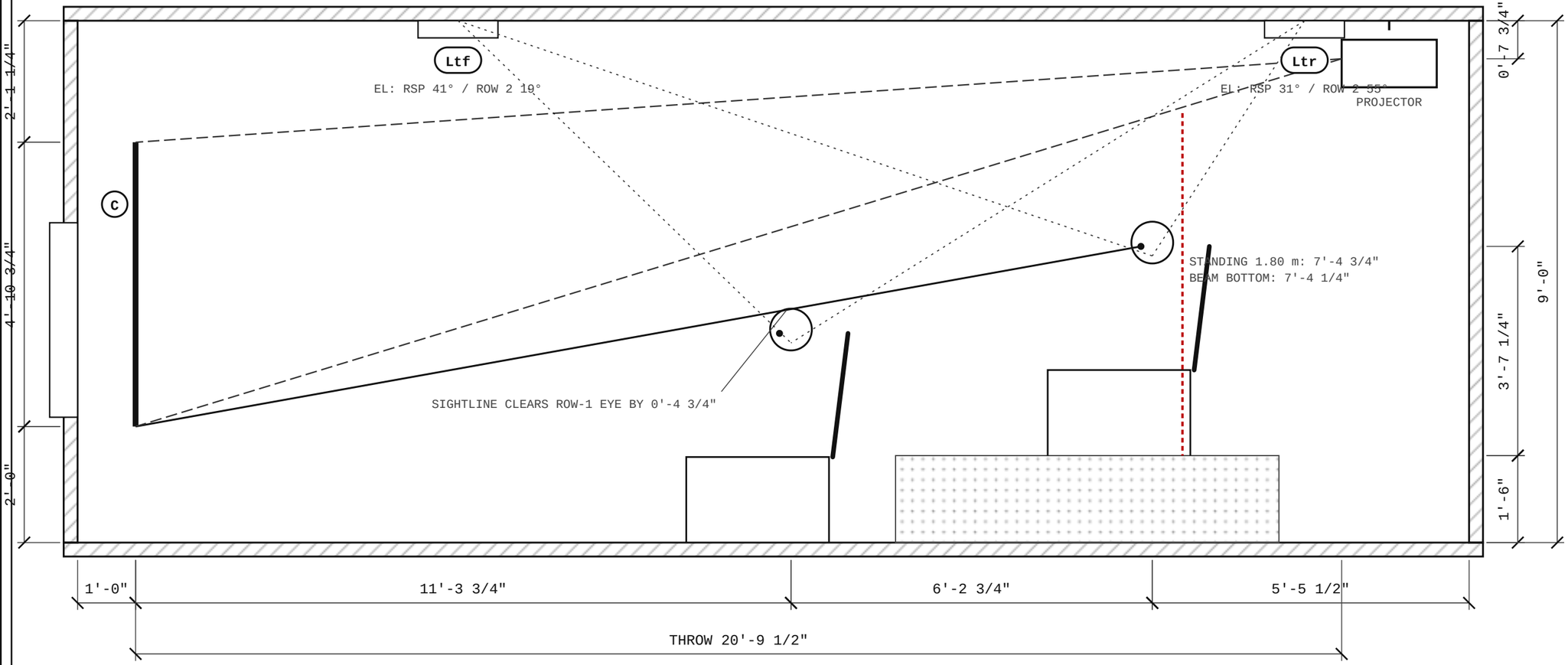
SCALE
3/8" = 1'-0" (on 11x17)

CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE
Side Wall Elevations

A-202



SECTION NOTES

1. Riser 1'-6" computed for a 120 mm sightline clearance (C-value) over row-1 eye to image bottom.
2. Viewing: row 1 41.6° horizontal / 16.3° to image top; row 2 27.7° / 5.8°.
3. Headroom on riser 7'-6". Overhead angles shown at RSP and row-2 centre (Dolby window 30-55°, verify).
4. Standing viewers on the riser intersect the lower image beam; consider a higher mount or rear-wall shelf (see AV-601).

CHT

CHICAGOLAND HOME THEATER

PROJECT
JBL Synthesis 24 x 18

PROJECT NO.
CHT-P-0002

ROOM
24'-0" L x 18'-0" W x 9'-0" H

FORMAT
Dolby Atmos 7.2.4 / 4 subwoofers

REVISION
B (concept, Savant-priced packages)

DATE
2026-11-09

DRAWN
CHT design engine v0.1

SCALE
1/2" = 1'-0" (on 11x17)

CONCEPT DESIGN

NOT FOR CONSTRUCTION

SHEET TITLE
Longitudinal Section

A-301

Schedules: loudspeakers, equipment, amplification, cabling

LOUDSPEAKER SCHEDULE

TAG	MODEL	MOUNTING	X (FROM LEFT WALL)	Y (FROM SCREEN WALL)	Z (A.F.F.)	AZIMUTH @RSP	ELEVATION @RSP	DISTANCE	AIM
L	JBL Synthesis SCL-2	Front wall, behind AT screen/fabric	2'-4"	0'-0"	3'-10"	28.4°	1.6°	14'-0"	flush, on-axis
C	JBL Synthesis SCL-2	Front wall, behind AT screen/fabric	9'-0"	0'-0"	3'-10"	0.0°	1.8°	12'-3 3/4"	flush, on-axis
R	JBL Synthesis SCL-2	Front wall, behind AT screen/fabric	15'-8"	0'-0"	3'-10"	28.4°	1.6°	14'-0"	flush, on-axis
Lss	JBL Synthesis SCL-3	Left wall	0'-0"	14'-1"	6'-2 3/4"	101.1°	16.9°	9'-7"	flush, on-axis
Rss	JBL Synthesis SCL-3	Right wall	18'-0"	14'-1"	6'-2 3/4"	101.1°	16.9°	9'-7"	flush, on-axis
Lrs	JBL Synthesis SCL-3	Left wall	0'-0"	20'-7 3/4"	6'-2 3/4"	132.8°	12.8°	12'-7"	flush, on-axis
Rrs	JBL Synthesis SCL-3	Right wall	18'-0"	20'-7 3/4"	6'-2 3/4"	132.8°	12.8°	12'-7"	flush, on-axis
Ltf	JBL Synthesis SCL-3	Ceiling	6'-4 1/2"	6'-6 3/4"	9'-0"	24.6°	41.3°	8'-5"	yaw 163° / tilt 27°
Rtf	JBL Synthesis SCL-3	Ceiling	11'-7 1/2"	6'-6 3/4"	9'-0"	24.6°	41.3°	8'-5"	yaw -163° / tilt 27°
Ltr	JBL Synthesis SCL-3	Ceiling	6'-4 1/2"	21'-2"	9'-0"	163.5°	31.0°	10'-9 1/4"	yaw 25° / tilt 37°
Rtr	JBL Synthesis SCL-3	Ceiling	11'-7 1/2"	21'-2"	9'-0"	163.5°	31.0°	10'-9 1/4"	yaw -25° / tilt 37°
SW1	JBL Synthesis SSW-3	Left wall, vertical	0'-0"	6'-0"	3'-3"	-	-	-	-
SW2	JBL Synthesis SSW-3	Left wall, vertical	0'-0"	18'-0"	3'-3"	-	-	-	-
SW3	JBL Synthesis SSW-3	Right wall, vertical	18'-0"	6'-0"	3'-3"	-	-	-	-
SW4	JBL Synthesis SSW-3	Right wall, vertical	18'-0"	18'-0"	3'-3"	-	-	-	-

Coordinates: origin at front-left floor corner at the finished faces; X across the room, Y from the screen wall, Z above finished floor. Azimuth 0° = toward the screen.

EQUIPMENT SCHEDULE (BETTER PACKAGE)

QTY	MODEL	SKU	DESCRIPTION	NOTE	LIST (MSRP)	EXT. LIST
3	SCL-2	JBLSCL-2	JBL Synthesis SCL-2 2.5-way in-wall LCR	LCR	·	·
8	SCL-3	SCL-3	JBL Synthesis SCL-3 dual 5.25in 2-way in-wall	surround + top	·	·
8	SCL-3PCB	SCL-3PCB	JBL Synthesis pre-construction bracket for SCL-3		·	·
4	SSW-3	JBLSSW3BLK	JBL Synthesis SSW-3 dual 10in passive in-wall subwoofer	Side walls at 1/4 and 3/4 length	·	·
1	SDP-60	JBLSDP60AM	JBL Synthesis SDP-60 16-ch AV processor		·	·
2	SMA-4750	JBLSMA4750AM	JBL Synthesis SMA-4750 4-ch Class D amplifier	7 channels used of 8	·	·
1	SMA-8300	JBLSMA8300AM	JBL Synthesis SMA-8300 8-ch Class D amplifier	8 channels used of 8	·	·
1	VPL-XW6100ES	VPL-XW6100ES	Sony VPL-XW6100ES 4K SXRD laser projector		·	·
1	K0701-0000	K0701-0000	Kaleidescape Strato V movie player		·	·
1	PKG-S2REM-41	PKG-S2REM-41	Savant S2 host with Pro Remote X2		·	·
1	ITP-E8000V3B-00	ITP-E8000V3B-00	Savant Touch 8 control screen, black		·	·
1	SEQ	SEQ	SurgeX sequencing surge eliminator 2RU 20 A		·	·
1	-	-	Acoustically transparent screen, 120in 16:9 fixed frame	price separately	-·	-·
8	-	-	Theater recliners	price separately	-·	-·
1	-	-	Speaker cable (12/14/10 AWG per AV-501 schedule)	price separately	-·	-·
1	-	-	Back boxes / brackets where JBL bracket not on Savant list	price separately	-·	-·
1	-	-	Acoustic treatment package (panels, cloud, bass traps, diffusion)	price separately	-·	-·
1	-	-	Equipment rack, ventilation, rack accessories	price separately	-·	-·
1	-	-	Labor: pre-wire, trim-out, calibration	price separately	-·	-·
			Equipment on Savant list			·

Processor outputs used: 13 of 16 (11 full-range + 2 sub outputs; 4 subwoofers). List prices: Savant integrator price list 20261103. Dealer cost is kept in the private parts catalog only.

AMPLIFIER CHANNEL MAP

CHANNEL	AMPLIFIER	OUTPUT
L	SMA-4750 #1	1
C	SMA-4750 #1	2
R	SMA-4750 #1	3
Lss	SMA-8300 #1	1
Rss	SMA-8300 #1	2
Lrs	SMA-8300 #1	3
Rrs	SMA-8300 #1	4
Ltf	SMA-8300 #1	5
Rtf	SMA-8300 #1	6
Ltr	SMA-8300 #1	7
Rtr	SMA-8300 #1	8
SW1	SMA-4750 #1	4
SW2	SMA-4750 #2	1
SW3	SMA-4750 #2	2
SW4	SMA-4750 #2	3

SPEAKER CABLE SCHEDULE

RUN	EST. LENGTH	GAUGE	LOOP R	LOSS
L	18.3 m (60 ft)	12 AWG	0.190 Ω	-0.27 dB
C	16.2 m (53 ft)	12 AWG	0.169 Ω	-0.24 dB
R	14.2 m (47 ft)	14 AWG	0.235 Ω	-0.33 dB
Lss	14.0 m (46 ft)	14 AWG	0.231 Ω	-0.33 dB
Rss	8.5 m (28 ft)	14 AWG	0.140 Ω	-0.20 dB
Lrs	12.0 m (39 ft)	14 AWG	0.198 Ω	-0.28 dB
Rrs	6.5 m (21 ft)	14 AWG	0.107 Ω	-0.15 dB
Ltf	13.5 m (44 ft)	14 AWG	0.223 Ω	-0.32 dB
Rtf	11.9 m (39 ft)	14 AWG	0.197 Ω	-0.28 dB
Ltr	9.0 m (30 ft)	14 AWG	0.149 Ω	-0.21 dB
Rtr	7.4 m (24 ft)	14 AWG	0.123 Ω	-0.18 dB
SW1	17.3 m (57 ft)	10 AWG	0.114 Ω	-0.24 dB
SW2	13.7 m (45 ft)	12 AWG	0.142 Ω	-0.30 dB
SW3	11.8 m (39 ft)	12 AWG	0.123 Ω	-0.26 dB
SW4	8.2 m (27 ft)	14 AWG	0.136 Ω	-0.29 dB

Rack outside the back wall. Length = vertical drop + ceiling route (Manhattan) + 1.5 m drop + 2.0 m service loop. Gauge chosen so loop resistance ≤ 5% of 0.8 × nominal impedance (stand-in for minimum impedance).

Performance calculations and design review

SPEAKER PLACEMENT COMPLIANCE PER SEAT

CH	WINDOW	1A	1B	1C	1D	2A	2B	2C	2D
L	az 22-30°	11°	23°	33°	42°	7°	16°	24°	30°
R	az 22-30°	42°	33°	23°	11°	30°	24°	16°	7°
Lss	az 90-110°	110°	103°	100°	98°	47°	60°	67°	71°
Rss	az 90-110°	98°	100°	103°	110°	71°	67°	60°	47°
Lrs	az 135-150°	150°	138°	129°	122°	114°	105°	101°	99°
Rrs	az 135-150°	122°	129°	138°	150°	99°	101°	105°	114°
Ltf	el 30-55°	43°	43°	38°	32°	19°	19°	18°	16°
Rtf	el 30-55°	32°	38°	43°	43°	16°	18°	19°	19°
Ltr	el 30-55°	32°	32°	30°	26°	53°	55°	40°	29°
Rtr	el 30-55°	26°	30°	32°	32°	29°	40°	55°	53°

Placement found by search over wall and ceiling positions (Phase 3, A4). 7.2.4: 38/80 seat-channel pairs in window (48%); 7.2.6: 42/96 (44%).
Selected 7.2.4. Windows: Dolby home guidelines as commonly cited [verify].

OUTPUT AND AMPLIFICATION (ESTIMATE)

CH	MODEL	AMP	AMP W @Z	DIST (M)	W FOR 105+3 DB	SPK MAX, FREE-FIELD	WITH AMP, +ROOM
L	SCL-2	SMA-4750	1000	4.27	967	104.6	108.0
C	SCL-2	SMA-4750	1000	3.75	748	105.7	108.6
R	SCL-2	SMA-4750	1000	4.27	967	104.6	108.0
Lss	SCL-3	SMA-8300	400	2.92	571	105.2	107.1
Rss	SCL-3	SMA-8300	400	2.92	571	105.2	107.1
Lrs	SCL-3	SMA-8300	400	3.83	983	102.8	105.8
Rrs	SCL-3	SMA-8300	400	3.83	983	102.8	105.8
Ltf	SCL-3	SMA-8300	400	2.56	439	106.3	107.9
Rtf	SCL-3	SMA-8300	400	2.56	439	106.3	107.9
Ltr	SCL-3	SMA-8300	400	3.29	722	104.2	106.5
Rtr	SCL-3	SMA-8300	400	3.29	722	104.2	106.5

Target 105 dB peaks per channel + 3 dB margin at the RSP. "+room" adds the statistical reverberant field (Q = 5, R = 155 m²). Class E estimate, about ±3 dB.

SUBWOOFER CONFIGURATIONS (MODAL MODEL)

SUBWOOFER LAYOUT	SUBS	FEASIBLE	MEAN SEAT SD (DB)	MAX SD (DB)	MAX LFE @RSP (DB)
Front wall pair, between C and L/R	2	yes	4.58	10.5	107.0
Front corners	2	yes	5.24	16.0	105.8
Rear corners	2	**no** (door)	4.28	9.5	106.1
Side-wall midpoints	2	yes	5.07	9.0	110.0
Front and back wall centres	2	**no** (C)	5.68	13.2	107.5
Diagonal corners (front-left + rear-right)	2	**no** (door)	4.50	8.4	105.9
Diagonal corners (front-right + rear-left)	2	yes	4.50	8.4	105.9
Four corners	4	**no** (door)	4.26	8.8	111.9
Four wall midpoints	4	**no** (C)	2.73	9.3	114.9
• Side walls at 1/4 and 3/4 length	4	yes	2.96	8.2	114.4
Front and back walls at 1/4 and 3/4 width	4	yes	3.54	9.8	113.0
Front pair (between C and L/R) + back wall pair	4	yes	4.62	10.2	113.2

Rigid-wall modal sum, T60 = 0.4 s below 250 Hz, 8 seats, 20-100 Hz. Ranks options; absolute levels need measurement.

ROOM ACOUSTICS

BAND (HZ)	RT60 UNTREATED (S)	RT60 TREATED (S)
125	0.42	0.42
250	0.73	0.33
500	0.52	0.20
1000	0.45	0.17
2000	0.40	0.16
4000	0.37	0.15

Axial modes below 120 Hz: 23.4, 31.3, 46.9, 62.5, 62.5, 70.3, 93.8, 93.8, 117.2 Hz. Modes below 100 Hz: 25. Small-room reference ≈ 0.26 s. Coefficients are illustrative typical values; replace with tested ISO 354 / ASTM C423 product data.

VIDEO

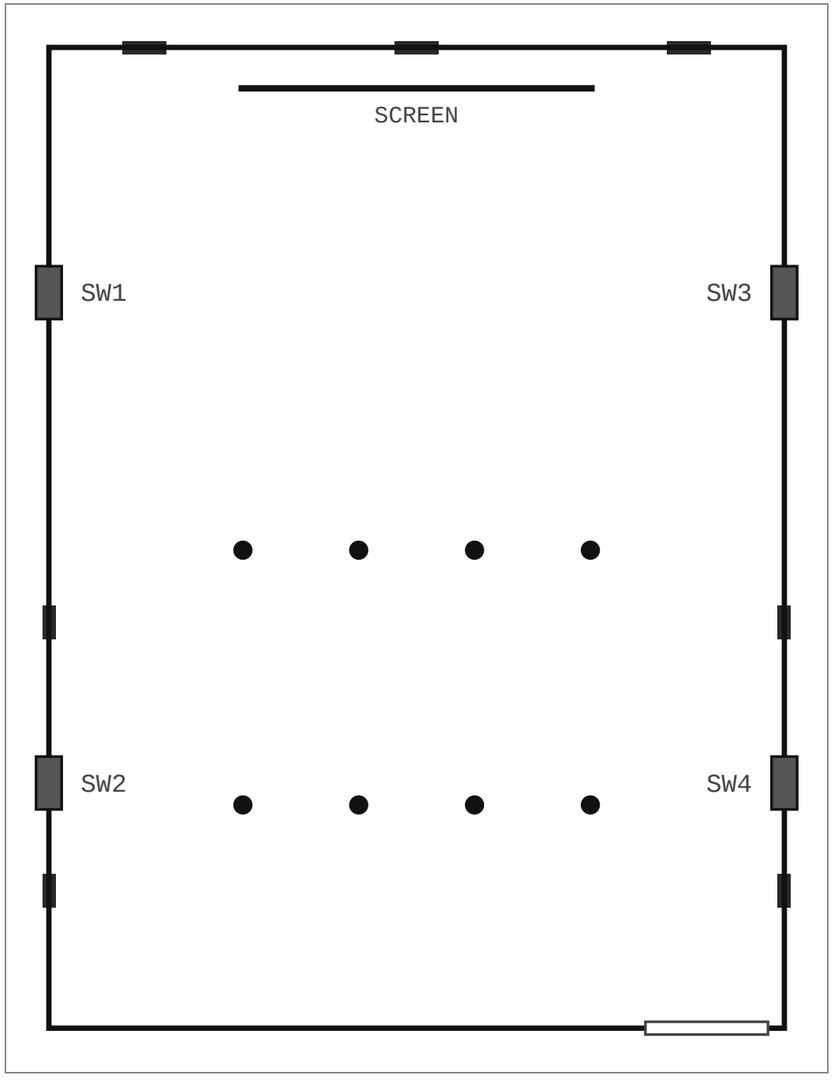
SEAT	TO SCREEN (M)	HORIZ. ANGLE	TO TOP	TO BOTTOM	OFF-CENTRE
1A	3.44	37.8°	16.3°	-8.1°	20.6°
1B	3.44	41.6°	16.3°	-8.1°	7.1°
1C	3.44	41.6°	16.3°	-8.1°	7.1°
1D	3.44	37.8°	16.3°	-8.1°	20.6°
2A	5.35	26.5°	5.8°	-10.0°	13.6°
2B	5.35	27.7°	5.8°	-10.0°	4.6°
2C	5.35	27.7°	5.8°	-10.0°	4.6°
2D	5.35	26.5°	5.8°	-10.0°	13.6°

Image 8'-8 1/2" × 4'-10 3/4" (42.7 ft²). Projector throw 20'-9 1/2", ratio 2.39:1, lens shift 79%. Luminance: k=0.65, gain 1.0: 41.1 fL (141 cd/m²); k=0.45, gain 1.0: 28.4 fL (97 cd/m²); k=0.65, gain 0.85: 34.9 fL (120 cd/m²). 4K pixel structure resolves inside 2.38 m.

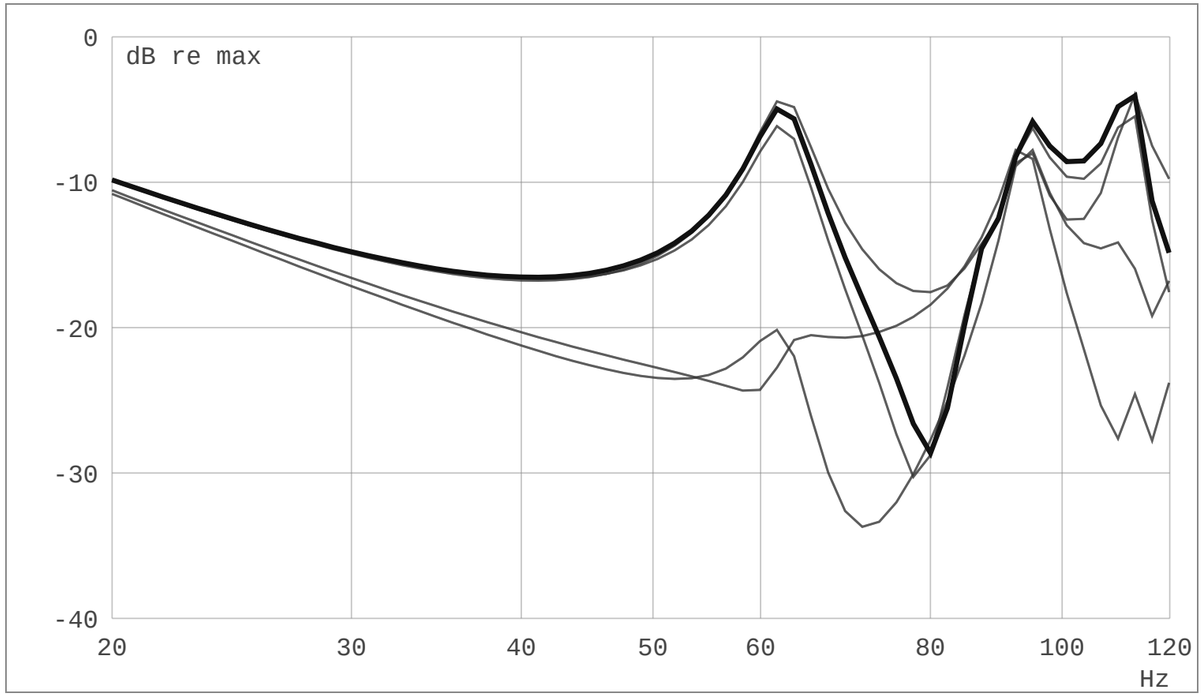
DESIGN REVIEW FLAGS

- Front-row vertical angle to image top 16.3 deg (>15 deg guidance [verify]).
- Standing 1.80 m person on riser reaches 2.26 m; beam bottom there 2.24 m → casts shadow when standing.
- Subwoofers (4 x SSW-3): est. free-field max 114.4 dB vs LFE target 115 dB; relies on low-frequency room gain (not modelled).
- Treated mid-band RT60 0.18 s is well below the ~0.26 s small-room reference: risk of an over-damped room. Trade some absorption for diffusion (e.g. rear half of side zones).
- 125 Hz RT60 0.42 s vs mid-band 0.18 s: add low-frequency absorption.
- Axial mode coincidence near 62.5 Hz: [0, 0, 1] and [2, 0, 0] (room ratios).
- Axial mode coincidence near 93.8 Hz: [0, 4, 0] and [3, 0, 0] (room ratios).
- Axial mode coincidence near 125.0 Hz: [0, 0, 2] and [4, 0, 0] (room ratios).

Subwoofer planner: SSW-3 layouts



MEAN SEAT SD	WORST-FREQUENCY SD
2.96 dB	8.2 dB
EST. MAX LFE AT RSP	4 X SSW-3
114.4 dB	Fits



Predicted low-frequency response at every seat (thin) and the RSP (bold), 20-120 Hz, rigid-wall modal model with T60 = 0.4 s. Use it to compare layouts; confirm with measurements. Mean seat SD = average across frequency of the level spread between the 8 seats (lower is more even).

Good / Better / Best options

PACKAGE COMPARISON

	GOOD	BETTER (DRAWN)	BEST
POSITIONING	Value JBL Synthesis in-wall system, AVR-based	Reference in-wall LCR, separates, four in-wall subs	SCL-1 compression-driver LCR, SSW-1 dual-15 subs, 36-ch processing
FRONT L / C / R	SCL-6	SCL-2	SCL-1
SURROUNDS	SCL-7	SCL-3	SCL-2
OVERHEADS	SCL-8	SCL-3	SCL-3
SUBWOOFERS	4 x SSW-4 (Side walls at 1/4 and 3/4 length)	4 x SSW-3 (Side walls at 1/4 and 3/4 length)	4 x SSW-1 (Side walls at 1/4 and 3/4 length)
PROCESSOR / OUTPUTS USED	SDR-38 / 13/16	SDP-60 / 13/16	SDP-80 / 13/36
AMPLIFICATION	1 x SMA-4750, 2 x SDA-1700	2 x SMA-4750, 1 x SMA-8300	3 x SMA-4750, 1 x SMA-8300
PROJECTOR	VPL-XW5100ES	VPL-XW6100ES	VPL-XW8100ES
EST. PEAKS AT RSP (+ROOM)	94.0-101.9 dB	105.8-108.6 dB	106.5-110.1 dB
EST. MAX LFE AT RSP	108.5 dB	114.4 dB	126.4 dB
BASS EVENNESS (MEAN SEAT SD)	2.77 dB	2.96 dB	2.90 dB
IMAGE BRIGHTNESS (EST.)	33 fL	41 fL	52 fL
DESIGN FLAGS	9	8	7
EQUIPMENT AT LIST (MSRP)	*use Show prices*	*use Show prices*	*use Show prices*

Same room, seating, screen and speaker positions for every package; products, subwoofer count and electronics change. Each package uses its own recommended subwoofer layout. List prices (MSRP) from the Savant integrator price list 20261103; they cover equipment on the Savant list only (screen, seating, wire, treatment and labor are priced separately). Dealer cost and margin are in the private CHT Parts Catalog workbook, not on this sheet. Regenerate the drawings for another package with: python3 design.py --tier good|better|best.

GOOD EQUIPMENT

QTY	ITEM	EXT. LIST
3	JBL Synthesis SCL-6 quad 5.25in in-wall compression driver	·
3	JBL Synthesis rough-in bracket for SCL-6	·
4	JBL Synthesis SCL-7 dual 5.25in angled in-wall	·
4	JBL Synthesis rough-in bracket for SCL-7	·
4	JBL Synthesis SCL-8 5.25in angled in-ceiling	·
4	JBL Synthesis SSW-4 dual 8in passive in-wall subwoofer	·
1	JBL Synthesis SDR-38 16-ch AV receiver (7 × 100 W)	·
1	JBL Synthesis SMA-4750 4-ch Class D amplifier	·
2	JBL Synthesis SDA-1700 2-ch subwoofer amplifier	·
1	Sony VPL-XW5100ES 4K SXRD laser projector	·
1	Kaleidescape Strato C movie player	·
1	Savant S2 host with Pro Remote X2	·
1	SurgeX rack surge eliminator 1RU 20 A	·
	Total at list	·

BETTER EQUIPMENT

QTY	ITEM	EXT. LIST
3	JBL Synthesis SCL-2 2.5-way in-wall LCR	·
8	JBL Synthesis SCL-3 dual 5.25in 2-way in-wall	·
8	JBL Synthesis pre-construction bracket for SCL-3	·
4	JBL Synthesis SSW-3 dual 10in passive in-wall subwoofer	·
1	JBL Synthesis SDP-60 16-ch AV processor	·
2	JBL Synthesis SMA-4750 4-ch Class D amplifier	·
1	JBL Synthesis SMA-8300 8-ch Class D amplifier	·
1	Sony VPL-XW6100ES 4K SXRD laser projector	·
1	Kaleidescape Strato V movie player	·
1	Savant S2 host with Pro Remote X2	·
1	Savant Touch 8 control screen, black	·
1	SurgeX sequencing surge eliminator 2RU 20 A	·
	Total at list	·

BEST EQUIPMENT

QTY	ITEM	EXT. LIST
3	JBL Synthesis SCL-1 dual 12in 2-way custom LCR	·
4	JBL Synthesis SCL-2 2.5-way in-wall LCR	·
4	JBL Synthesis SCL-3 dual 5.25in 2-way in-wall	·
4	JBL Synthesis pre-construction bracket for SCL-3	·
4	JBL Synthesis SSW-1 dual 15in passive subwoofer	·
1	JBL Synthesis SDP-80 36-ch preamp/processor	·
3	JBL Synthesis SMA-4750 4-ch Class D amplifier	·
1	JBL Synthesis SMA-8300 8-ch Class D amplifier	·
1	Sony VPL-XW8100ES 4K SXRD laser projector	·
1	Kaleidescape Strato K movie player	·
1	Kaleidescape Compact Terra movie server 30 TB	·
1	Savant S2 host with Pro Remote X2	·
1	Savant Touch 8 control screen, black	·
1	SurgeX sequencing surge eliminator 2RU 20 A	·
	Total at list	·

Budget builder

CATEGORY	INCLUDES	ESTIMATE	LOW	HIGH	SHARE
Audio	Loudspeakers, subwoofers, brackets, processor, amplification	$86,956	$78,260	$95,651	
Video	Projector, movie server / player, plus screen allowance (allowance not entered)	$22,995	$20,695	$25,294	
Control	Savant host, remote, touch screen	$3,300	$2,970	$3,630	
Power	Surge protection and power conditioning	$2,169	$1,952	$2,386	
Seating	Theater recliners (allowance per seat)	enter allowance			
Acoustic treatment	Panels, cloud, bass traps, diffusion (allowance)	enter allowance			
Infrastructure	Speaker wire, rack, ventilation, conduit (allowance)	enter allowance			
Labor & calibration	Pre-wire, trim-out, programming, calibration (allowance)	enter allowance			
Total	**Better package, ±10%**	**$115,420**	**$103,878**	**$126,962**	

Equipment estimates are list prices (MSRP) from the Savant integrator price list 20261103 for the selected package. Seating, screen, treatment, infrastructure and labor are not on the Savant list: enter your allowances below and they are added to their category. Low / High apply the range to every category. Allowances are remembered in this browser only.

ALLOWANCES (NOT ON THE SAVANT LIST)

AT screen, 120in 16:9 fixed frame
$ enter allowance

Recliner, per seat (x 8 seats)
$ enter allowance

Acoustic treatment package
$ enter allowance

Wire, rack and infrastructure
$ enter allowance

Labor, programming and calibration
$ enter allowance

3D model and client spin



Drag to orbit, scroll or pinch to zoom. The screen always shows the Chicagoland Home Theater logo; load your own logo file (PNG or SVG) to replace the wordmark. It is kept in this browser only. With Show prices on, each part is labelled with its model and list price. Exports save after you confirm. Labels are not part of the exported video.
